Supplementary notes to *Background Expectation for the LZ High-Energy Event*

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These notes collect material that was developed in the course of the analysis and is not needed to follow its argument. None of it bears on the central result, which is that the published characterisation of the electron-recoil response does not determine the tail probability on which the background expectation depends. We record it because each item either answers a question a reader may reasonably ask of the main text, or documents a calculation whose conclusion is quoted there without its derivation. Section and equation numbers of the main text are referred to as they appear there.

S1 Contours of constant N_q in the observable plane

Recombination conserves N_q, so it displaces an event along a contour of constant total quanta. Since $\mathrm{S1c} = g_1 N_{ph}$ and $\mathrm{S2c} = g_2 N_e$, that contour is

$$\frac{\mathrm{S1c}}{g_1} + \frac{\mathrm{S2c}}{g_2} = N_q \qquad \Longleftrightarrow \qquad \mathrm{S2c} = g_2\left(N_q - \frac{\mathrm{S1c}}{g_1}\right), \tag{S1}$$

a straight line of slope $-g_2/g_1 = -312.5$ in the $\{\mathrm{S1c}, \mathrm{S2c}\}$ plane. In the $\{\mathrm{S1c}, \log_{10}\mathrm{S2c}\}$ plane in which the bands are drawn, the same locus is $\log_{10}[g_2(N_q - \mathrm{S1c}/g_1)]$, a curve whose local slope

$$\frac{\mathrm{d}\log_{10}\mathrm{S2c}}{\mathrm{d}\,\mathrm{S1c}} = -\frac{g_2/g_1}{\ln 10\ \mathrm{S2c}} \tag{S2}$$

steepens from -0.0032 dex/phd at the nominal recombination of a parent of 69.84 $\mathrm{keV_{ee}}$to -0.0148 dex/phd at the event, and diverges as $\mathrm{S1c} \to g_1 N_q$. These loci are the contours of constant energy drawn in grey in Figs. 2 and 4 of the Letter, where their curvature is directly visible.

The distinction matters when a vertical separation is read off the band. The bands of the Letter are evaluated taking flat energy spectra along lines of fixed S1c, so the reference population at fixed S1c spans many parent energies: at $\mathrm{S1c} = 540$ phd the band median corresponds to a parent of 89.8 $\mathrm{keV_{ee}}$rather than the 69.84 $\mathrm{keV_{ee}}$of the event inversion. A vertical displacement therefore mixes recombination with parent energy, whereas a displacement along eq. (S1) isolates recombination at fixed N_q. The inference of the Letter is a two-dimensional unbinned profile likelihood ratio and uses neither projection; we record the distinction because the vertical separation is quoted elsewhere as a measure of separation from electron-recoil backgrounds.

S2 The low-energy edge of the acceptance

The main text adopts 0.62 $\mathrm{keV_{ee}}$for the lower edge of the electron-recoil acceptance of the WS RoI and states that three plausible treatments span 2.2% in the acceptance width. The derivation is as follows.

The ceiling is obtained by solving $Q_y(E)\,E = 409.5$ with the tuned charge-yield model, giving 20.02 $\mathrm{keV_{ee}}$at $Q_y = 20.46$ $\mathrm{e^-/keV}$. The floor cannot be obtained that way, for two reasons.

First, the yield parameterisation is not usable at sub-keV energies. Charge yield is bounded above by the case in which no ion recombines,

$$Q_y^{\max} = \frac{1}{W\,(1 + N_{ex}/N_i)} = 62.6\ \mathrm{e^-/keV}, \tag{S3}$$

whereas the tuned model tends to $C\,m_2 = 85.3$ $\mathrm{e^-/keV}$ as $E \to 0$, that is $m_2 = 84.9$ $\mathrm{e^-/keV}$ before the density correction $C = 1.0051$, exceeding the ceiling by 36%. The two cross at $E = 0.97$ $\mathrm{keV_{ee}}$:

E [keV]	$Q_y(E)$	$N_e = Q_y E$	$N_i = E/[W(1 + N_{ex}/N_i)]$	$N_e > N_i$?
0.193	84.6	16.3	12.1	yes
0.26	82.9	21.6	16.3	yes
0.50	78.0	39.0	31.3	yes
1.0	70.6	70.6	62.6	marginal
5.0	44.1	220	313	no
20.0	20.5	409	1251	no
70.0	17.7	1237	4379	no

Below about 1 $\mathrm{keV_{ee}}$the parameterisation returns more electrons than there are ions, and is being used outside the range over which it was tuned.

Second, the WS RoI is three cuts rather than one. Taking them together, S1c > 3 phd requires $N_{ph} > 27.2$ and S2 > 645 phd requires $N_e > 18.7$ — the latter binding more tightly than the S2c floor's 16.3 — so $N_q > 45.9$ and $E > 0.62$ $\mathrm{keV_{ee}}$. At that N_q one has $N_i = 38.9$ and $N_{ex} = 7.0$, and the two requirements can be met only for $r \simeq 0.52$, whereas the capped model gives full ionisation there and S1c = 0.78 phd. Sub-keV electron recoils therefore enter the RoI only on a recombination fraction well above the value the capped model returns, and 0.62 $\mathrm{keV_{ee}}$is a boundary of the acceptance rather than a threshold that typical events cross.

treatment	floor [$\mathrm{keV_{ee}}$]	width [$\mathrm{keV_{ee}}$]
S2c cut alone, uncapped yield	0.193	19.80
S2c cut alone, full ionisation	0.261	19.73
all three cuts jointly	0.621	19.37

The spread in $\mathrm{d}N/\mathrm{d}E$ across the three is 2.2%. The first is listed only to show what the parameterisation returns if its sub-keV extrapolation is taken at face value; discarding it leaves 1.9%.

S3 Bounds on the width-variation sensitivity

The main text notes that a width known to better than +6.2% would keep the within-family contribution below a tenth of the quoted background, and that +16.6% is where it reaches the whole of it. The corresponding bounds, obtained by folding the varied response through the parent spectrum, are

if σ_{N_e} is high by at most	$N_{\mathrm{leak}}^{\mathrm{ER}}$ below	fraction of 0.0106
+2%	3.4×10^{-4}	3%
+5%	7.8×10^{-4}	7%
+10%	2.7×10^{-3}	25%
+16%	9.5×10^{-3}	90%

Only upward variations are tabulated; downward variations reduce the leakage, so these are the conservative side. These bounds constrain the within-family sensitivity alone. They do not bear on the identifiability result of the main text, every member of whose families carries the same σ_{N_e}.

S4 The double-vacancy populations

The Letter identifies the ^{124}Xe and ^{125}I double-vacancy decays as the populations whose projection into the tail carries a systematic it does not propagate. We examined both externally. An external reconstruction of these populations rests on extrapolations that no published measurement constrains, and what follows establishes two bounds and one non-identifiability rather than a determination of the response. That is why it is recorded here and not in the main text.

These decays produce more light and less charge than equivalent-energy β decays because the energy is distributed among multiple low-energy Auger–Meitner electrons, raising the ionisation density and hence recombination [1].

^{124}Xe. With $T_{1/2} = 1.09 \times 10^{22}$ yr and a KK fraction of 64.8% from the collaboration's own measurement [2], the exposure contains 787 2νDEC decays, of which 510 are KK captures at 64.3 keV. The rate calculation can be checked against the same paper's statement that 2.3% of the signal lies below the region of interest of that analysis: that fraction gives 18 events against the 21.0 ± 6.3 of Table I of the Letter.

The KK mode is not strongly charge-suppressed. In 88% of K-captures a 28 keV X-ray carries energy $\mathcal{O}(100)$ μm from the primary site [1], so the deposit comprises three independent recombination sites. Only the both-Auger channel, $(1 - f_K)^2 \simeq 1.6\%$ or ~ 7 events, with f_K the K-shell fluorescence yield, is a single dense deposit; with the measured box-size ratio $a_{\mathrm{L}}/a_\beta = 0.84 \pm 0.03$ in the Thomas–Imel framework [1, 3] this gives $Q_y/Q_y^\beta \geq 0.774$ and a contribution of 5×10^{-5} counts. This contribution is negligible.

Taking $a_{\mathrm{KK}} \geq a_{\mathrm{L}}$ is the conservative choice. Scaling the box size with the electron range would give substantially larger suppression and is not supported by the data: the measured $a_{\mathrm{L}}/a_\beta = 0.84$ corresponds to a range ratio of 0.10, so the box size is far less sensitive to track structure than the range. The KK double-Auger cascade is moreover dominated by $\sim$23 keV KLL electrons, which are more β-like than the sub-keV L cascade.

^{125}I. The mechanism is the Letter's own: its Fig. 4 caption records that the spectral features in the ER band above 25 keV$_{\mathrm{ee}}$come predominantly from ^{125}I and ^{133}Xe and appear slightly below the β-decay ER band due to increased recombination from Auger cascades.

^{125}I decays by electron capture to ^{125}Te, followed by a 35.49 keV nuclear transition internally converted in 93% of decays. That energy lies just above the xenon K edge at 34.56 keV [4], where the mean free path is $\simeq 0.14$ mm, so it deposits locally in essentially every decay and produces a second vacancy; the capture-shell label refers to the first vacancy only. The K, L and M modes deposit 67.3, 40.4 and 36.5 keV, all above the 20.02 keV$_{\mathrm{ee}}$at which a mean-recombination electron recoil meets the S2c ceiling. The 8.9 ± 2.7 ^{125}I events of Table I therefore cannot be ordinary electron recoils of the corresponding energies: they are a population whose *mean* charge yield is suppressed.

shell	Q_y/Q_y^β	$\langle N_e\rangle$	σ_{N_e}	$(\langle N_e\rangle - 409.5)/\sigma_{N_e}$	$P(N_e \le 409)$
K	0.773	916	194	2.61	8.3×10^{-4}
L	0.783	562	99	1.54	2.8×10^{-2}
M	0.785	517	87	1.24	7.7×10^{-2}

Here $\langle N_e\rangle = (Q_y/Q_y^\beta)\, Q_y(E)\, E$ is evaluated at each mode's own energy, and σ_{N_e} is the full width including the binomial term, with ω taken from Eq. 4 of the Supplement at that N_q: it is not a constant, falling from 0.0456 at $N_q = 4973$ to 0.0369 at $N_q = 2697$. The implied recombination fractions, $1 - \langle N_e\rangle/N_i$, are 0.783, 0.778 and 0.774; these are not the Q_y/Q_y^β ratios, which they resemble numerically and order in the opposite sense, and each exceeds the nominal $\bar{r}$ at that energy by the double-vacancy enhancement. The Gaussian-equivalent displacements $\Phi^{-1}(1-P)$ are 3.1, 1.9 and 1.4, larger than $(\langle N_e\rangle - 409.5)/\sigma_{N_e}$ because the distribution is positively skewed and its low-side tail correspondingly thinner than Gaussian.

It is important not to overstate this. The L and M modes reach the WS RoI through 1.9 and 1.4 Gaussian-equivalent standard deviations, probabilities of order 10^{-2} to 10^{-1}; they are not a probe of the $P \simeq 10^{-7}$ regime that governs the ordinary electron-recoil leakage. What places them in the search region is the shift of the mean, not a rare fluctuation. Only the K mode, at 3.1, approaches the displacement at issue, and only its S1c interval brackets the event at 540.1 phd: S1c $>$ 504 phd follows from $N_e \le 409$, and S1c $<$ 549 phd from $N_e \ge 0$, that is S1c $\le N_q g_1$. The L and M modes, at $N_q = 2988$ and 2697, land at S1c $\simeq$ 285–330 and 253–298 phd, inside the WS RoI but well outside the comparison region. Weighted by capture fraction they supply $\sim$92% of the ^{125}I population in the WS RoI and the K mode the remaining 8% — not because the K mode is absent from its S1c range, but because reaching the ceiling from a mean of 916 demands a much larger downward fluctuation than the L and M modes require from 562 and 517. The implied total is $\sim 10^3$ decays.

The size of the K-mode population is not recoverable from outside, and the sensitivity runs parallel to the identifiability result of the main text. Assigning the L mode the single-vacancy L-capture ratio measured in Ref. [1] (0.877) rather than the double-vacancy treatment moves the fraction above S1c = 504 phd from 8% to 82%, and the inferred total ^{125}I decay count by an order of magnitude. Neither assignment is a measurement of the ^{125}I response. In the absence of a measured double-vacancy response distribution, modelling choices available externally change the inferred population in the event's S1c region by a factor of ten — which is the regime in which Ref. [1] states that the Thomas–Imel framework cannot predict the size of the effect a priori.

S5 An attempted reconstruction of the published band

Figure 2 of the Letter draws the electron-recoil band as "the calibrated NEST model, taking flat energy spectra and evaluating along lines of fixed S1c". We implemented that construction, in order to express the deviations of section S1 against the published band rather than against the response at a single energy, and did not succeed. We record the attempt because the main text's quantities are defined at fixed energy partly in consequence.

At fixed S1c the photon number is fixed, and since recombination conserves quanta, $N_q = N_{ph} + N_e$. For a flat energy spectrum the constraint collapses the energy integral exactly,

$$p(N_e \,|\, N_{ph}) \;\propto\; \int \mathrm{d}E\; f_E(N_e)\, \delta\!\left(\tfrac{E}{W} - N_{ph} - N_e\right) \;=\; W\, f_{E^*}(N_e), \qquad E^* = (N_{ph} + N_e)\, W, \quad \text{(S4)}$$

so each N_e is evaluated at the one energy that makes the quanta add up.

The reconstruction reproduces the published *median* to better than 0.011 in $\log_{10}$ S2c at every S1c, and two independent digitisations of the figure agree with one another to ~ 0.005. The *width* does not reproduce:

S1c [phd]	E at the median [keV$_{ee}$]	published 10–90% [dex]	reconstructed	ratio
100	17	0.134	0.148	0.91
200	33	0.159	0.177	0.90
300	48	0.172	0.214	0.81
500	82	0.230	0.291	0.79
540	90	0.246	0.301	0.82

The disagreement is about 10% at 17 keV$_{ee}$and 20% above 48, with the step falling where the electron-recoil calibration of Ref. [5] ends. Two candidate causes sit inside our own assumptions and we cannot separate them from public information. The range of the "flat energy spectra" is not stated; truncating it at $\sim$100 keV$_{ee}$reproduces the published width at S1c = 540 almost exactly, 0.250 against 0.246, while changing nothing below S1c = 400, so a residual 10% persists at low S1c that the truncation does not touch. Effects we omit — Fano smearing of N_q and the binomial draw of N_i — would widen our band further and so cannot account for the gap, but the unknown truncation narrows it, and no claim can be made in either direction.

Two things this does *not* indicate. The reconstruction of the response model itself is verified independently: across the 24 scan energies it reproduces the N_q, mean N_e and standard deviation that NEST returns to 0.002%, 0.003% and 0.54%, and the variance and skew-normal placement match the source line for line. And the apparent symmetry of the published band about its median is an artefact of the logarithmic axis: in N_e both bands are positively skewed, with $(N_{90} - N_{50})/(N_{50} - N_{10})$ rising with S1c in each, so the skewness treatment is not a separate discrepancy from the width.

The scripts, the digitised curves and a fuller account are in the archive, as `analysis/band.py` and `notes/band-reconstruction.md`.

References

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